

## Module 2

# Accounting Statements and Finance

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FINE 3010, Financial Management  
Prof. Renping Li, Tulane University

# Why Do We Need Accounting?

1. What do firms own?
  - Real assets (tangible, intangible) and financial assets (cash, securities)

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1. What do firms own?
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2. What do firms owe?

# Why Do We Need Accounting?

1. What do firms own?
  - Real assets (tangible, intangible) and financial assets (cash, securities)
2. What do firms owe?
3. How much money did firms make?
  - Income? Cash flow?

## Example, Is Anthropic Profitable?

- Is Anthropic profitable in Q2 2026?



CNBC, August 18, 2026 (3:50),  
“Anthropic tells investors annualized  
revenue run rate climbed to \$65  
billion in July”

## Example, Is Anthropic Profitable?

- Is Anthropic profitable in Q2 2026?
- We will have a clear understanding after this module



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# Three Financial Statements

- **Balance sheet**
  - What do firms own and owe at a point in time



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- **Balance sheet**
  - What do firms own and owe at a point in time
- **Income statement**
  - How much did firms earn in the past year



# Three Financial Statements

- **Balance sheet**
  - What do firms own and owe at a point in time
- **Income statement**
  - How much did firms earn in the past year
- **Statement of cash flows**
  - Cash inflows and outflows during the past year



# Securities and Exchange Commission (SEC)

- **SEC**, the federal agency that regulates U.S. securities markets and enforces disclosure



[sec.gov/edgar/search](https://sec.gov/edgar/search)

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# Goals

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- Interpret the balance sheet, income statement, and statement of cash flows
- Distinguish between market and book values
- Explain why income differs from cash flow
- Understand the taxation of corporate and personal income

## Module 2, Part 1

# The Balance Sheet

# Balance Sheet

- **Balance sheet**, assets, liabilities, and equity at a point in time

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$$\text{Assets} = \text{Liabilities} + \text{Equity}$$

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*Where does the balance sheet come from?*

- Recall from **Module 1**, businesses sell financial assets (financing) to purchase real assets (investment)

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- The numbers stay at the transaction amount, even when the asset is worth more or less later

## Example, Issuing Stock to Build a Data Center (\$ Billions)

### 1. Today

|                     |          |                                     |          |
|---------------------|----------|-------------------------------------|----------|
| Cash                | 1        | Liabilities                         | 0        |
| Fixed assets        | 0        | Shareholders' equity                | 1        |
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### 2. Issue stock

|                     |          |                                     |          |
|---------------------|----------|-------------------------------------|----------|
| Cash                | 2        | Liabilities                         | 0        |
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| 3. Build data center |          |                                     |          |
|----------------------|----------|-------------------------------------|----------|
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- *Should you revise fixed assets if the data center is now worth more than \$2 billion?*

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- *Should you revise fixed assets if the data center is now worth more than \$2 billion? **No***
- *Should you revise shareholders' equity if someone is willing to buy the business at \$4 billion?*

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- *Should you revise shareholders' equity if someone is willing to buy the business at \$4 billion? **No***

# Balance Sheet

*Where does the balance sheet come from?*

- Recall from **Module 1**, businesses sell financial assets (financing) to purchase real assets (investment)
- The balance sheet “grows” in this process
- The numbers are the cost of acquiring real assets, or the amount raised from selling financial assets
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# Breaking Down the Balance Sheet of a Business (\$ Millions)

## Current assets

Assets that can generate return within one year

### Assets

#### Current assets

|                                |        |
|--------------------------------|--------|
| Cash and marketable securities | 2,600  |
| Accounts receivable            | 500    |
| Inventories                    | 9,000  |
| Total current assets           | 12,100 |

#### Fixed assets

##### Tangible fixed assets

|                               |        |
|-------------------------------|--------|
| Property, plant, equipment    | 48,200 |
| Less accumulated depreciation | 19,700 |
| Net tangible fixed assets     | 28,500 |

|                  |     |
|------------------|-----|
| Intangible asset | 700 |
|------------------|-----|

|                    |        |
|--------------------|--------|
| Total fixed assets | 29,200 |
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|                     |               |
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| <b>Total assets</b> | <b>41,300</b> |
|---------------------|---------------|

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1. Cash and marketable securities

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# Breaking Down the Balance Sheet of a Business (\$ Millions)

## Current assets

Assets that can generate return within one year

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2. Accounts receivable
  - Amount owed by customers

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- Property, plant, equipment ("PP&E"), the amount in the table is always the acquisition cost
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# Breaking Down the Balance Sheet of a Business (\$ Millions)

## Current liabilities

Obligations due within a year

### Liabilities and equity

#### Current liabilities

|                        |     |
|------------------------|-----|
| Debt due for repayment | 200 |
|------------------------|-----|

|                  |       |
|------------------|-------|
| Accounts payable | 9,900 |
|------------------|-------|

|                           |        |
|---------------------------|--------|
| Total current liabilities | 10,100 |
|---------------------------|--------|

|                |        |
|----------------|--------|
| Long-term debt | 11,300 |
|----------------|--------|

|                   |        |
|-------------------|--------|
| Total liabilities | 21,400 |
|-------------------|--------|

#### Shareholders' equity

|                 |       |
|-----------------|-------|
| Paid-in capital | 6,300 |
|-----------------|-------|

|                   |        |
|-------------------|--------|
| Retained earnings | 13,600 |
|-------------------|--------|

|                            |        |
|----------------------------|--------|
| Total shareholders' equity | 19,900 |
|----------------------------|--------|

|                                     |               |
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| <b>Total liabilities and equity</b> | <b>41,300</b> |
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# Breaking Down the Balance Sheet of a Business (\$ Millions)

## Current liabilities

Obligations due within a year

1. Debt due for repayment
  - Debt due for repayment within a year

## Liabilities and equity

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# Breaking Down the Balance Sheet of a Business (\$ Millions)

## Current liabilities

Obligations due within a year

1. Debt due for repayment
  - Debt due for repayment within a year
2. Accounts payable
  - Amount owed to suppliers

## Liabilities and equity

Current liabilities

|                        |     |
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|-------------------|--------|

Shareholders' equity

|                 |       |
|-----------------|-------|
| Paid-in capital | 6,300 |
|-----------------|-------|

|                   |        |
|-------------------|--------|
| Retained earnings | 13,600 |
|-------------------|--------|

|                            |        |
|----------------------------|--------|
| Total shareholders' equity | 19,900 |
|----------------------------|--------|

|                                     |               |
|-------------------------------------|---------------|
| <b>Total liabilities and equity</b> | <b>41,300</b> |
|-------------------------------------|---------------|

# Breaking Down the Balance Sheet of a Business (\$ Millions)

## Current liabilities

Obligations due within a year

1. Debt due for repayment
  - Debt due for repayment within a year
2. Accounts payable
  - Amount owed to suppliers

## Liabilities and equity

|                                     |               |
|-------------------------------------|---------------|
| Current liabilities                 |               |
| Debt due for repayment              | 200           |
| Accounts payable                    | 9,900         |
| <b>Total current liabilities</b>    | <b>10,100</b> |
| Long-term debt                      | 11,300        |
| Total liabilities                   | 21,400        |
| Shareholders' equity                |               |
| Paid-in capital                     | 6,300         |
| Retained earnings                   | 13,600        |
| Total shareholders' equity          | 19,900        |
| <b>Total liabilities and equity</b> | <b>41,300</b> |

# Breaking Down the Balance Sheet of a Business (\$ Millions)

## Long-term liabilities

Obligations due after a year

### Liabilities and equity

#### Current liabilities

|                        |     |
|------------------------|-----|
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|------------------------|-----|

|                  |       |
|------------------|-------|
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# Breaking Down the Balance Sheet of a Business (\$ Millions)

## Long-term liabilities

Obligations due after a year

### 1. Long-term debt

- Bonds, debt from public bond markets
- Bank loans, debt from banks

## Liabilities and equity

### Current liabilities

|                        |     |
|------------------------|-----|
| Debt due for repayment | 200 |
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# Breaking Down the Balance Sheet of a Business (\$ Millions)

## Shareholders' equity

What is left to its owners, a *residual* term

## Liabilities and equity

|                                     |                      |
|-------------------------------------|----------------------|
| Current liabilities                 |                      |
| Debt due for repayment              | 200                  |
| Accounts payable                    | 9,900                |
| Total current liabilities           | <u>10,100</u>        |
| Long-term debt                      | 11,300               |
| Total liabilities                   | <u>21,400</u>        |
| Shareholders' equity                |                      |
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# Breaking Down the Balance Sheet of a Business (\$ Millions)

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What is left to its owners, a *residual* term

1. Paid-in capital
  - Amount raised from previous stock issuances

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## Shareholders' equity

What is left to its owners, a *residual* term

1. Paid-in capital
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  - Net income that the business has retained and reinvested on the shareholders' behalf

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|                                     |                      |
|-------------------------------------|----------------------|
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|-------------------------------------|----------------------|

# Breaking Down the Balance Sheet of a Business (\$ Millions)

| Most liquid<br> | Assets                     |               | Liabilities and equity              |               | Paid first<br> |
|--------------------------------------------------------------------------------------------------|----------------------------|---------------|-------------------------------------|---------------|---------------------------------------------------------------------------------------------------|
|                                                                                                  | Current assets             |               | Current liabilities                 |               |                                                                                                   |
|                                                                                                  | Cash and marketable sec.   | 2,600         | Debt due for repayment              | 200           |                                                                                                   |
|                                                                                                  | Accounts receivable        | 500           | Accounts payable                    | 9,900         |                                                                                                   |
|                                                                                                  | Inventories                | 9,000         | Total current liabilities           | 10,100        |                                                                                                   |
|                                                                                                  | Total current assets       | 12,100        | Long-term debt                      | 11,300        |                                                                                                   |
|                                                                                                  | Fixed assets               |               | Total liabilities                   |               |                                                                                                   |
|                                                                                                  | Tangible fixed assets      |               |                                     |               |                                                                                                   |
|                                                                                                  | Property, plant, equipment | 48,200        | Shareholders' equity                |               |                                                                                                   |
|                                                                                                  | Less accum. depreciation   | 19,700        | Paid-in capital                     | 6,300         |                                                                                                   |
| Least liquid                                                                                     | Net tangible fixed assets  | 28,500        | Retained earnings                   | 13,600        | Paid last                                                                                         |
|                                                                                                  | Intangible asset           | 700           | Total shareholders' equity          | 19,900        |                                                                                                   |
|                                                                                                  | Total fixed assets         | 29,200        |                                     |               |                                                                                                   |
|                                                                                                  | <b>Total assets</b>        | <b>41,300</b> | <b>Total liabilities and equity</b> | <b>41,300</b> |                                                                                                   |

- **Liquidity**, the ease of converting an asset into cash

# Book Value vs. Market Value

1. **Book values**, the cost of acquiring real assets, or the amount raised from selling financial assets
  - Values on balance sheets are book value

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  - E.g., worth \$4 billion today → market value

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  2. **Market values**, what things would sell for today
    - E.g., worth \$4 billion today → market value
- In general, book  $\neq$  market value

## Example, Issuing Stock to Build a Data Center (\$ Billions)

| 1. Today            |          |                                     |          |
|---------------------|----------|-------------------------------------|----------|
| Cash                | 1        | Liabilities                         | 0        |
| Fixed assets        | 0        | Shareholders' equity                | 1        |
| <b>Total assets</b> | <b>1</b> | <b>Total liabilities and equity</b> | <b>1</b> |

| 2. Issue stock      |          |                                     |          |
|---------------------|----------|-------------------------------------|----------|
| Cash                | 2        | Liabilities                         | 0        |
| Fixed assets        | 0        | Shareholders' equity                | 2        |
| <b>Total assets</b> | <b>2</b> | <b>Total liabilities and equity</b> | <b>2</b> |

| 3. Build data center |          |                                     |          |
|----------------------|----------|-------------------------------------|----------|
| Cash                 | 0        | Liabilities                         | 0        |
| Fixed assets         | 2        | Shareholders' equity                | 2        |
| <b>Total assets</b>  | <b>2</b> | <b>Total liabilities and equity</b> | <b>2</b> |

- *Should you revise fixed assets if the data center is now worth more than \$2 billion? **No***
- *Should you revise shareholders' equity if someone is willing to buy the business at \$4 billion? **No***

# Book Value vs. Market Value

1. Book value of equity = Book value of assets – Book value of liabilities
  - E.g., Apple at the end of fiscal year 2025,
    - Book value of assets, \$359,241 Mn
    - Book value of liabilities, \$285,508 Mn
    - What is the book value of equity?

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    - Answer,  $\$359,241 \text{ Mn} - \$285,508 \text{ Mn} = \$73,733 \text{ Mn}$

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    - Answer,  $\$359,241 \text{ Mn} - \$285,508 \text{ Mn} = \$73,733 \text{ Mn}$
2. Market value of equity = Market price of each share  $\times$  Total shares outstanding
  - E.g., Apple at the end of 2025,
    - Share price, about \$271
    - Shares outstanding, 14,800 Mn
    - What is the market value of equity?

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  - E.g., Apple at the end of 2025,
    - Share price, about \$271
    - Shares outstanding, 14,800 Mn
    - What is the market value of equity?
    - Answer,  $\$271 \times 14,800 \text{ Mn} = \$4,010,800 \text{ Mn}$

# Book Value vs. Market Value

- Book value of equity reflects the money that equity investors have invested into the firm



# Book Value vs. Market Value

- Book value of equity reflects the money that equity investors have invested into the firm
- Market value of equity reflects the money investors expect the firm to make in the future



# Book Value vs. Market Value

- Book value of equity reflects the money that equity investors have invested into the firm
- Market value of equity reflects the money investors expect the firm to make in the future
- Naturally, market value of equity  $>$  book value of equity for a successful business



## Practice Problem

- A business's book value of total assets is \$45,000. Its book value of current liabilities is \$15,000, and its book value of shareholders' equity is \$20,000. How much is its book value of long-term liabilities?

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- $\text{Assets} = \text{Liabilities} + \text{Equity}$
- $\$45,000 = \$15,000 + \text{_____} + \$20,000$
- $\$45,000 = \$15,000 + \$10,000 + \$20,000$

# Recap of Part 1

- Accounting conveys information to investors and stakeholders
- The balance sheet, assets, liabilities, and equity at a point in time
- $\text{Assets} = \text{Liabilities} + \text{Equity}$
- In general, book  $\neq$  market value

## Module 2, Part 2

# The Income Statement

# Income Statement

- **Income statement**, how much firms earned in the past year



INCOME STATEMENTS

# Income Statement

- **Revenues**, sales made in the period, whether or not cash was collected
  - E.g., a firm sells 100 products to customers, each at a price of \$100
  - $\text{Revenues} = 100 \times \$100 = \$10,000$

|                               | \$ Million    |
|-------------------------------|---------------|
| <b>Revenues</b>               | <b>80,000</b> |
| COGS                          | 56,000        |
| SG&A expenses                 | 16,000        |
| Depreciation                  | 3,000         |
| <b>EBIT</b>                   | <b>5,000</b>  |
| Interest expense              | 1,000         |
| <b>Taxable income</b>         | <b>4,000</b>  |
| Taxes                         | 840           |
| <b>Net income</b>             | <b>3,160</b>  |
| Allocation of net income      |               |
| Dividends                     | 1,000         |
| Addition to retained earnings | 2,160         |

# Income Statement

- **Operating expenses**, the costs of running the business

|                               | \$ Million   |
|-------------------------------|--------------|
| Revenues                      | 80,000       |
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| Allocation of net income      |              |
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# Income Statement

- **Operating expenses**, the costs of running the business
  - **Cost of goods sold (COGS)**, the direct cost of the goods actually sold
    - E.g., each of the 100 products sold costs the firm \$60 to make
    - $\text{COGS} = 100 \times \$60 = \$6,000$

|                               | \$ Million    |
|-------------------------------|---------------|
| Revenues                      | 80,000        |
| <b>COGS</b>                   | <b>56,000</b> |
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  - **Selling, general & administrative (SG&A)**, other operating costs, e.g., marketing, salaries, distribution
  - **Depreciation**, this period's share of a fixed asset's cost

|                               | \$ Million   |
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# Fixed Assets Spread over Time as Depreciation

- Fixed assets earn revenue for years, so we spread their cost over those years instead of subtracting it all at once



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- Fixed assets earn revenue for years, so we spread their cost over those years instead of subtracting it all at once
- This spreading is **depreciation**
- Two things happen with depreciation,
  - In the income statement, subtract depreciation from revenues as a non-cash expense



# Fixed Assets Spread over Time as Depreciation

- Fixed assets earn revenue for years, so we spread their cost over those years instead of subtracting it all at once
- This spreading is **depreciation**
- Two things happen with depreciation,
  - In the income statement, subtract depreciation from revenues as a non-cash expense
  - In the **balance sheet**, add to accumulated depreciation



# Fixed Assets Spread over Time as Depreciation

- The firm pays \$2 billion to build a data center that lasts 5 years

# Fixed Assets Spread over Time as Depreciation

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## Fixed Assets Spread over Time as Depreciation

- The firm pays \$2 billion to build a data center that lasts 5 years
- We spread the cost over these 5 years
- Each year, the depreciation is \$0.4 billion

| Year                    | 0    | 1    | 2    | 3    | 4    | 5    |                |
|-------------------------|------|------|------|------|------|------|----------------|
| Cash flow (\$ billions) | -2.0 | 0    | 0    | 0    | 0    | 0    |                |
| Effect on net income    | 0    | -0.4 | -0.4 | -0.4 | -0.4 | -0.4 | ← Depreciation |

# Income Statement

- **EBIT**, earnings before interest and taxes
  - $\text{EBIT} = \text{Revenues} - \text{COGS} - \text{SG\&A expenses} - \text{Depreciation}$

|                               | \$ Million   |
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| Allocation of net income      |              |
| Dividends                     | 1,000        |
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# Income Statement

- **Interest expense**, the interest paid on the firm's debt for the period

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| Allocation of net income      |              |
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# Income Statement

- **Taxes**

- Taxable income = EBIT – Interest expense
- Taxes = Taxable income × Tax rate

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| <b>Net income</b>             | <b>3,160</b> |
| Allocation of net income      |              |
| Dividends                     | 1,000        |
| Addition to retained earnings | 2,160        |

## Where Net Income Goes

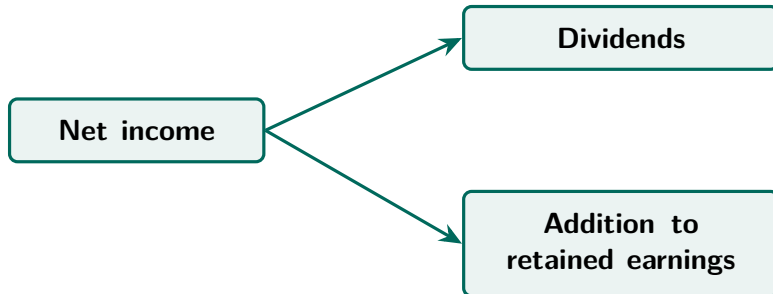
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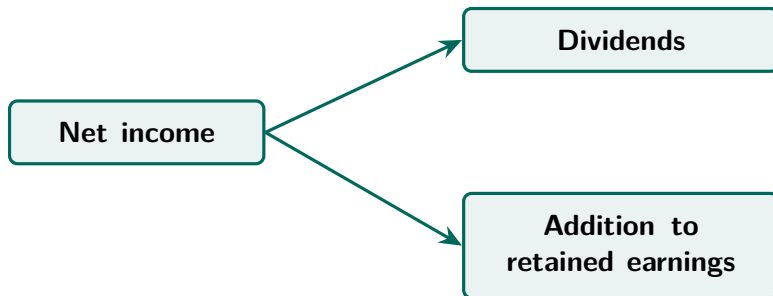
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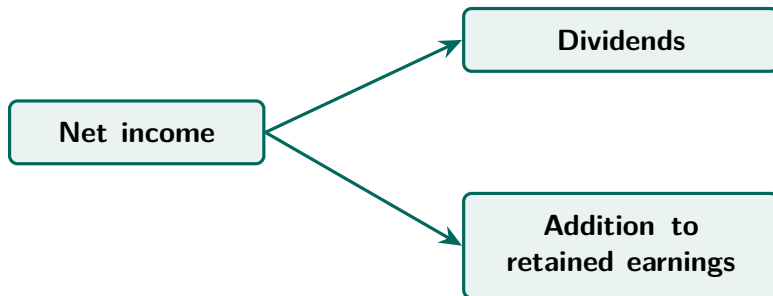
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- **Net income**, what the business earned in the period after every expense
- Shareholders have claims to net income
- Net income can be kept in the business, or paid out to shareholders



- **Dividends**, cash paid out to shareholders now
- **Addition to retained earnings**, profit kept in the business and reinvested on the shareholders' behalf

# Income Statement, A Memory Aid

| Line item                     | Letters | Remember it as                     |
|-------------------------------|---------|------------------------------------|
| Revenues                      | R       | <b>R</b> ainbow                    |
| COGS                          | C       | <b>C</b> andy                      |
| SG&A expenses                 | S       | <b>S</b> ells                      |
| Depreciation                  | D       | <b>D</b> ependably                 |
| EBIT                          | E       | <b>E</b> specially                 |
| Interest expense              | I       | <b>I</b> n                         |
| Taxable income                | T.I.    | <b>T</b> exas <b>I</b> ncorporated |
| Taxes                         | T       | <b>T</b> exas                      |
| Net income                    | N.I.    | <b>N</b> ever <b>I</b> ncorporates |
| Dividends                     | D       | <b>D</b> uring                     |
| Addition to retained earnings | R.E.    | <b>R</b> e- <b>E</b> lections      |

## Example, Is Anthropic Profitable?

- Is Anthropic profitable in Q2 2026?



CNBC, August 18, 2026 (3:50),  
“Anthropic tells investors annualized  
revenue run rate climbed to \$65  
billion in July”

## Example, Is Anthropic Profitable?

- Is Anthropic profitable in Q2 2026?
- Profitable on an **EBITDA** basis
  - EBITDA, earnings before interest, taxes, depreciation, and amortization
  - Amortization, the same cost-spreading as depreciation, applied to intangible assets



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- Not profitable on a **net income** basis



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## Practice Problem

|                               | \$            |
|-------------------------------|---------------|
| Revenues                      | ?             |
| COGS                          | 20,000        |
| SG&A expenses                 | 10,000        |
| Depreciation                  | 8,000         |
| <b>EBIT</b>                   | <b>12,000</b> |
| Interest expense              | ?             |
| <b>Taxable income</b>         | <b>?</b>      |
| Taxes (tax rate = 21%)        | 2,100         |
| <b>Net income</b>             | <b>7,900</b>  |
| Dividends                     | 2,500         |
| Addition to retained earnings | ?             |

## Practice Problem

|                               | \$            |
|-------------------------------|---------------|
| Revenues                      | 50,000        |
| COGS                          | 20,000        |
| SG&A expenses                 | 10,000        |
| Depreciation                  | 8,000         |
| <b>EBIT</b>                   | <b>12,000</b> |
| Interest expense              | 2,000         |
| <b>Taxable income</b>         | <b>10,000</b> |
| Taxes (tax rate = 21%)        | 2,100         |
| <b>Net income</b>             | <b>7,900</b>  |
| Dividends                     | 2,500         |
| Addition to retained earnings | 5,400         |

## Recap of Part 2

- The income statement, how much firms earned in the past year
- **Depreciation**, this period's share of a fixed asset's cost
- Net income can be kept in the business, or paid out to shareholders
- Taxes are computed **after** depreciation and interest, never before

## Module 2, Part 3

# The Statement of Cash Flows

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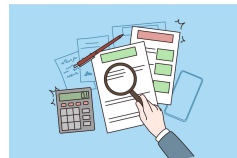
# Statement of Cash Flows

| Direct method                        | Indirect method                       |
|--------------------------------------|---------------------------------------|
| Lists every cash receipt and payment | Starts from net income and adjusts it |
| More time-consuming to prepare       | Faster to prepare                     |
| Rarely used in practice              | <b>The common choice</b>              |

Direct →



Indirect →



# Statement of Cash Flows

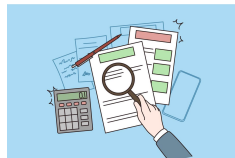
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|--------------------------------------|---------------------------------------|
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- Both methods give the same totals

Direct →



Indirect →



# Statement of Cash Flows

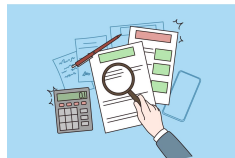
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| More time-consuming to prepare       | Faster to prepare                     |
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- Both methods give the same totals
- We will follow the indirect method, starting from net income in the income statement

Direct →



Indirect →



# How Do Net Income and Cash Flow Differ?

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| Year                    | 0    | 1    | 2    | 3    | 4    | 5    |
|-------------------------|------|------|------|------|------|------|
| Cash flow (\$ billions) | -2.0 | 0    | 0    | 0    | 0    | 0    |
| Effect on net income    | 0    | -0.4 | -0.4 | -0.4 | -0.4 | -0.4 |

← Depreciation

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| Effect on net income    | 0    | -0.4 | -0.4 | -0.4 | -0.4 | -0.4 |

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- Cash leaves all at once, net income feels it slowly

*Relationship of cash flow (CF) and net income (NI)?*

*When spending on fixed assets,  $CF = NI - \text{spending on fixed assets}$*

*When recording depreciation,  $CF = NI + \text{depreciation}$*

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*When raising money,  $CF = NI + \text{amount raised}$*

# How Do Net Income and Cash Flow Differ?

- The firm spends \$60 million producing AI control panels
- None of them have been sold yet



OpenAI, July 15, 2026 (2:15),  
“Introducing the Codex Micro”

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|                              | Effect     |
|------------------------------|------------|
| Cash flow (\$ millions)      | −60        |
| Effect on net income         | 0          |
| <u>Change in inventories</u> | <u>+60</u> |



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*When adding to inventories,  $CF = NI - \text{change in inventories}$*



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***When firm not yet paid by customers,  $CF = NI - \text{change in accounts receivable}$***

# How Do Net Income and Cash Flow Differ?

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|                                   | Effect     |
|-----------------------------------|------------|
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***When firm pays off suppliers,  $CF = NI + \text{change in accounts payable}$***



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# Operating Activities

**Operating activities**, cash from running the business day to day

Steps to calculate cash flow from operating activities,

1. **Start** with net income
2. **Add** depreciation
3. **Subtract** change in net working capital (NWC)
  - $\text{NWC} = \text{inventories} + \text{accounts receivable} - \text{accounts payable}$
  - NWC, cash tied up in day-to-day operations, in unsold goods and unpaid bills

# Why Subtract Change in Net Working Capital (NWC)?

*When adding to inventories,  $CF = NI - \text{change in inventories}$*

*When firm not yet paid by customers,  $CF = NI - \text{change in accounts receivable}$*

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$$CF = NI$$

- change in inventories*
- change in accounts receivable*
- + change in accounts payable*

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*Ignoring other adjustments,*

$$CF = NI$$

*– change in inventories*

*– change in accounts receivable*

*+ change in accounts payable*

$$CF = NI - \text{change in (inventories + accounts receivable – accounts payable)}$$

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*Ignoring other adjustments,*

$$CF = NI$$

*– change in inventories*

*– change in accounts receivable*

*+ change in accounts payable*

$$CF = NI - \text{change in (inventories + accounts receivable – accounts payable)}$$

$$CF = NI - \text{change in net working capital (NWC)}$$

# Operating Activities

| Cash flow from operating activities |               |
|-------------------------------------|---------------|
| Net income                          | 3,160         |
| Depreciation                        | 3,000         |
| Changes in working capital items    | <u>-1,000</u> |
| Cash flow from operating activities | <u>7,160</u>  |

$$3,160 + 3,000 - (-1,000) = 7,160$$

# Investing Activities

**Investing activities**, cash spent buying fixed assets and cash received from selling them

Investing activities show cash flows from,

1. Capital purchased and sold

- **Capital**, a firm's fixed assets
- e.g., machinery, plant, land
- **Capital expenditure**, the amount spent on buying capital during the period
- Cash flow is *negative* if firms buy capital

# Investing Activities

| Cash flow from investing activities |               |
|-------------------------------------|---------------|
| Capital expenditure                 | −3,000        |
| Sales of fixed assets               | 200           |
| Cash flow from investing activities | <u>−2,800</u> |

$$-3,000 + 200 = -2,800$$

# Financing Activities

**Financing activities**, cash raised from lenders and shareholders, and cash returned to them

Financing activities show cash flows from,

1. Selling new shares → Cash ↑
2. Paying dividends → Cash ↓
3. Taking new debt, either a loan or a bond → Cash ↑
4. Repaying principal amount of old debt → Cash ↓

## Recap of Part 3

- The statement of cash flows, cash receipts and payments over a period
- Net income  $\neq$  cash flow
- Cash flow from operating activities = Net income + Depreciation – Change in NWC
- **Investing activities**, cash spent buying fixed assets and cash received from selling them
- **Financing activities**, cash raised from lenders and shareholders, and cash returned to them

## Practice Problem

- Which of the following will make cash flow higher than net income for a firm?
  - a. Increase in inventories.
  - b. Decrease in accounts payable.
  - c. Decrease in accounts receivable.
  - d. Expenditures on fixed assets.

## Practice Problem

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## Practice Problem

- A business has net income of \$3,160 million and depreciation of \$3,000 million, and its NWC changed by  $-\$1,000$  million. What is its cash flow from operating activities?

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## Practice Problem

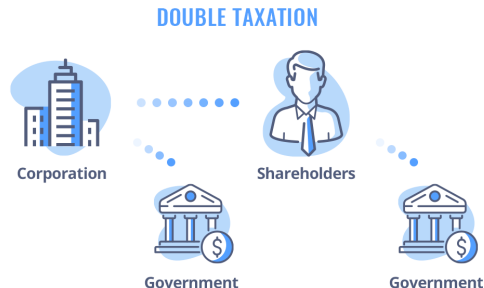
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- Cash flow from operating activities = Net income + Depreciation – Change in NWC
- $\$3,160 + \$3,000 - (-\$1,000) = \$7,160$  million

## Module 2, Part 4

# Taxation

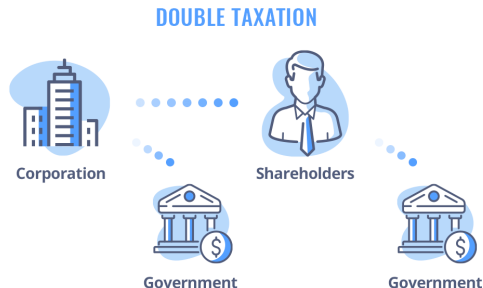
# Corporate Taxation, Double Taxation

- Recall from **Module 1**,
  - The corporation pays tax on its income



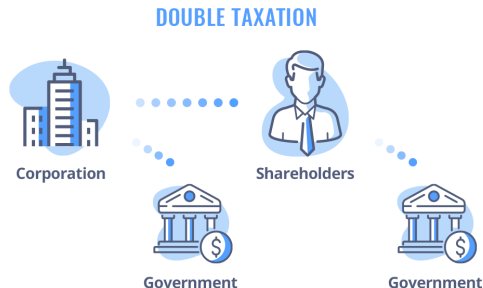
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  - Shareholders pay tax again on what they receive



# Corporate Taxation, Double Taxation

- Recall from **Module 1**,
  - The corporation pays tax on its income
  - Shareholders pay tax again on what they receive
- This is **double taxation**



# Corporate Taxation, The Tax Rate

- The U.S. federal corporate tax rate is **21%**



WSJ, February 28, 2025 (6:57),  
“Economists on How Trump’s 2017  
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- The rate is a major political topic
  - Higher rate, smaller deficit, more funding for public services



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- **TCJA (2017)**, cut the rate from 35% to 21%
- **One Big Beautiful Bill Act (2025)**, extended and made permanent much of it



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“Economists on How Trump’s 2017  
Tax Cuts Actually Played Out”

# Tax Shields

- Depreciation and interest come first

| Line item                     | Letters | Remember it as                     |
|-------------------------------|---------|------------------------------------|
| Revenues                      | R       | <b>R</b> ainbow                    |
| COGS                          | C       | <b>C</b> andy                      |
| SG&A expenses                 | S       | <b>S</b> ells                      |
| Depreciation                  | D       | <b>D</b> ependably                 |
| EBIT                          | E       | <b>E</b> specially                 |
| Interest expense              | I       | <b>I</b> n                         |
| Taxable income                | T.I.    | <b>T</b> exas <b>I</b> ncorporated |
| Taxes                         | T       | <b>T</b> exas                      |
| Net income                    | N.I.    | <b>N</b> ever <b>I</b> ncorporates |
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- Depreciation and interest reduce corporate taxes

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# Interest Tax Shield

- Interest expense can
  - act as a “shield” against corporate taxation
  - increase the *total* cash flow to investors (debt and equity investors combined)
- Investors get \$4,160 in total with debt, \$3,950 without

## A. Has debt

|                        |                |                    |
|------------------------|----------------|--------------------|
| <b>EBIT</b>            | <b>\$5,000</b> |                    |
| Interest expense       | \$1,000        | → debt investors   |
| <b>Taxable income</b>  | <b>\$4,000</b> |                    |
| Taxes (tax rate = 21%) | \$840          |                    |
| <b>Net income</b>      | <b>\$3,160</b> | → equity investors |

## B. No debt

|                        |                |                    |
|------------------------|----------------|--------------------|
| <b>EBIT</b>            | <b>\$5,000</b> |                    |
| Interest expense       | \$0            | → debt investors   |
| <b>Taxable income</b>  | <b>\$5,000</b> |                    |
| Taxes (tax rate = 21%) | \$1,050        |                    |
| <b>Net income</b>      | <b>\$3,950</b> | → equity investors |

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- Should a firm take on as much debt as possible? No, bankruptcy costs.

# Depreciation Tax Shield

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  - act as a “shield” against corporate taxation
  - increase the *total* cash flow to investors (debt and equity investors combined)

## Income Statement (\$ Million)

| Baseline              |              |
|-----------------------|--------------|
| Revenues              | 80,000       |
| COGS                  | 56,000       |
| SG&A expenses         | 16,000       |
| Depreciation          | 3,000        |
| <b>EBIT</b>           | <b>5,000</b> |
| Interest expense      | 1,000        |
| <b>Taxable income</b> | <b>4,000</b> |
| Taxes                 | 840          |
| <b>Net income</b>     | <b>3,160</b> |

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|                               | Baseline     | More depreciation |
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| COGS                          | 56,000       | 56,000            |
| SG&A expenses                 | 16,000       | 16,000            |
| Depreciation                  | 3,000        | 4,000             |
| <b>EBIT</b>                   | <b>5,000</b> | <b>4,000</b>      |
| Interest expense              | 1,000        | 1,000             |
| <b>Taxable income</b>         | <b>4,000</b> | <b>3,000</b>      |
| Taxes                         | 840          | 630               |
| <b>Net income</b>             | <b>3,160</b> | <b>2,370</b>      |

# Depreciation Tax Shield

| \$ Million            | Baseline     | More depreciation |
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| Net income            | 3,160        | 2,370             |
| Add back depreciation | 3,000        | 4,000             |
| <b>Cash flow</b>      | <b>6,160</b> | <b>6,370</b>      |

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- Net income *falls* by \$790 million
- Recall Part 3, Cash flow from operating activities = Net income + Depreciation – Change in NWC
- Cash flow *rises* by \$210 million
- Depreciation reduces taxable income, cuts taxes, and raises cash flow
  - Cash flow increases by Tax rate  $\times$  \$1 if depreciation is \$1 higher

## Practice Problem

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## Practice Problem

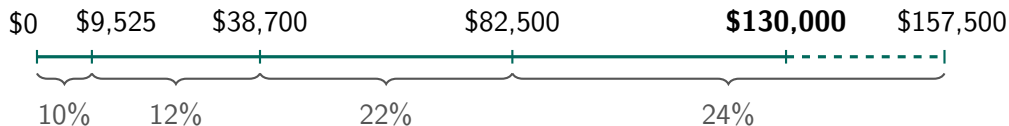
- A business pays a 21% tax rate. Suppose it had an additional \$200 million in depreciation. By how much would its taxes fall? What about its cash flow?
- Taxes fall by  $21\% \times \$200 \text{ million} = \$42 \text{ million}$
- Cash flow rises by the same \$42 million, the depreciation tax shield

# Personal Taxation

- Personal tax rates matter for salaried employees and non-corporate businesses
- Marginal tax rate = Tax rate that is applied to each extra dollar of income
- Average tax rate =  $\text{Total taxes} / \text{Total income}$

# Personal Taxation

| Taxable Income (Single Taxpayers) | Marginal Tax Rate (%) |
|-----------------------------------|-----------------------|
| \$0–\$9,525                       | 10                    |
| \$9,526–\$38,700                  | 12                    |
| \$38,701–\$82,500                 | 22                    |
| \$82,501–\$157,500                | 24                    |
| \$157,501–\$200,000               | 32                    |
| \$200,001–\$500,000               | 35                    |
| \$500,001 and above               | 37                    |



## Example, Average Tax Rate

Suppose your salary is \$130,000 per year. What would be your average tax rate?

| Taxable Income (Single Taxpayers) | Marginal Tax Rate (%) |
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| \$157,501–\$200,000               | 32                    |
| \$200,001–\$500,000               | 35                    |
| \$500,001 and above               | 37                    |

- Total tax =  $\$9,525 \times 10\% + \$ (38,700 - 9,525) \times 12\% + \$ (82,500 - 38,700) \times 22\% + \$ (130,000 - 82,500) \times 24\% = \$25,489.5$
- Average tax rate =  $25,489.5 / 130,000 = 19.61\%$

## Practice Problem

If a firm's net income is positive and its depreciation is positive, which of the following could account for a negative cash flow from operating activities?

- a. Inventories decrease while accounts receivable and accounts payable stay the same.
- b. Accounts receivable increases while inventories and accounts payable stay the same.
- c. The firm raises money by issuing stocks.
- d. A large addition is made to plant and equipment.

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## Recap of Part 4

- The U.S. federal corporate tax rate is **21%**
- Interest and depreciation come before taxes, so both shield income from tax
- A tax shield is worth the tax rate times the deduction, leaving more cash for investors
- Marginal rate applies to each extra dollar of income, average rate is total taxes over total income

# Key Takeaways

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- Three statements, what a firm owns and owes, what it earned, and how cash moved
- Book values are the cost of acquiring real assets or the amount raised from selling financial assets, market values are what things would sell for today
- Net income  $\neq$  cash flow, add back depreciation, subtract the change in NWC
- Interest and depreciation shield taxes, leaving more cash for investors

# Thank you!

Please let me know if you have any questions.